Computer Systems

4th September 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Where markers award half marks in any part of a question, they should ensure that the total mark recorded for the question is rounded up to a whole mark.

Answer ALL questions

Marks

Question 1

a) Select ONE (1) option that represents a collection of computer peripherals. **1**

i) CPU, RAM, Motherboard
ii) Keyboard, Mouse, Monitor
iii) Operating System, Application Software, Utility Software
iv) Headphones, Mouse, Internet

Mark Scheme:
ii) Keyboard, Mouse, Monitor ***(1 mark)***

b) Select ONE (1) option that is a function of an Operating System. **1**

i) Protecting the user from spyware
ii) Running application software
iii) Designing user interfaces
iv) Compilation of code

Mark Scheme:
ii) Compilation of code ***(1 mark)***

c) Structured Query Language (SQL) is a language for manipulating and querying databases. Select ONE (1) option that is **NOT** a valid SQL statement. **1**

i) SELECT
ii) AMEND
iii) UPDATE
iv) DELETE

Mark Scheme:
ii) AMEND ***(1 mark)***

d) Which type of memory is considered volatile? Select ONE (1) option. **1**

i) ROM (Read-Only Memory)
ii) Flash Memory
iii) RAM (Random Access Memory)
iv) SSD (Solid State Drive)

Mark Scheme:
iii) RAM (Random Access Memory) ***(1 mark)***

e) What is the binary representation of the hexadecimal value CF. Select ONE (1) option. **1**

i) 00111010
ii) 10100011
iii) 00110101
iv) 11001111

Mark Scheme:
iv) 11001111 ***(1 mark)***

f) Identify FIVE (5) key components that make up a Central Processing Unit (CPU). **5**

i) Arithmetic Logic Unit (ALU)
ii) Control Unit (CU)
iii) Register File
iv) Graphics Processing Unit (GPU)
v) Memory Management Unit (MMU)
vi) Floating Point Unit (FPU)
vii) Instruction Decoder
viii) Cache Memory

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Mark Scheme:
i)) Arithmetic Logic Unit (ALU)
ii) Control Unit (CU)
viii) Cache Memory
iii) Register File
vi) Floating Point Unit (FPU)

Total 10 Marks

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Question 2

You have been asked to build a Personal Computer (PC) for a close friend. Your friend would like to use the PC for word processing and browsing the internet. They already have a keyboard, mouse, and a monitor.

a) Name FIVE (5) *main computing hardware* components would you need to create a functioning PC? **5**

Mark Scheme:
*You would need to source a CPU **(1 mark)** or RAM **(1 mark)** or PSU **(1 mark)** or Hard Drive/Solid-State Drive **(1 mark)** or graphics card **(1 mark)** or Motherboard **(1 mark).***

b) Your friend has informed you that the computer struggles to play games on the PC. A quick diagnosis shows the screen stuttering when attempting to play visually demanding games. **1**

Identify the internal hardware component that you would suggest they upgrade, to ensure visually demanding games run smoothly on the PC?

Mark Scheme:
*Graphics Card **OR** Video Interface Card **(1 mark)***

c) Which component(s) of the PC would require updating if you started to run out of space for permanently storing documents? **1**

Mark Scheme:
*Hard Drive **OR** Solid-State Drive **(1 mark)***

d) The Central Processing Unit (CPU) contains a small amount of cache memory. Explain the purpose of cache memory. **3**

Mark Scheme:

*Cache memory in the CPU serves as a high-speed buffer between the CPU and the slower main memory (RAM) **(1 mark).** Its purpose is to store frequently accessed data and instructions from the main memory **(1 mark)**, reducing the time needed to fetch them during subsequent operations **(1 mark)**.*

Total 10 Marks

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Question 3

a) Computer systems process binary numbers which can be expressed in different formats.

i) Add binary numbers 10101 and 00100. Give the result as a binary number. **1**

Mark Scheme:

11001 ***(1 mark)***

ii) Convert the numbers from i) to decimal to prove your answer. Show your working. **3**

Mark Scheme:

16	***8***	***4***	***2***	***1***	***Decimal***	
1	***0***	***1***	***0***	***1***	***16 + 4 + 1 = 21***	***(1 mark for conversion)***
0	***0***	***1***	***0***	***0***	***4***	***(1 mark for conversion)***
			Total:		***21 + 4 = 25 (11001)***	***(1 mark for proof)***

iii) Convert the hexadecimal number F8D4 to its decimal value. Show your working. **2**

Mark Scheme:

F = 15

8 = 8

d = 13

4 = 4

$(15 \times 16^3) + (8 \times 16^2) + (13 \times 16^1) + (4 \times 16^0) = 63700$ ***(1 mark for correct answer, 1 mark for correct working)***

iv) Use two's complement method to negate the binary number 11010011. **2**

Mark Scheme:
00101100 ***(1 mark for inverting)***
00101100 + 1 = 00101101 ***(1 mark for adding 1 and correct result)***

v) What is the octal equivalent of the hexadecimal value B35? Show your working. **2**

Mark Scheme:
Hex to Binary
B = 1011
3 = 0011
5 = 0101

Binary to Octal
101 – 100 – 110 – 101 ***(1 mark for evidence of working)***

Binary to Decimal
(101 = 5, 100 = 4, 110 = 6, 101 = 5) = 5465^8 ***(1 mark for correct result)***

Total 10 Marks

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Question 4

The Operating System (OS) is a crucial part of any computer system. It serves as the interface between the user and the components.

a) In addition to the OS providing a user interface, outline and describe THREE (3) additional purposes of the OS? **9**

Mark Scheme

Any 3 from:

Resource Management (1 mark) – *Allocates and manages hardware resources* ***(1 mark) LIKE*** *CPU* ***OR*** *memory* ***OR*** *storage devices* ***OR*** *peripherals* ***(1 mark)***

Process Management (1 mark) - *Oversees process execution OR scheduling tasks OR managing process states* ***(1 mark) AND*** *facilitating inter-process communication and synchronisation* ***(1 mark)***

Memory Management (1 mark) - *Controls memory allocation to processes, manages virtual memory* ***(1 mark) AND*** *optimises memory usage by swapping data between RAM and secondary storage devices* ***(1 mark)***

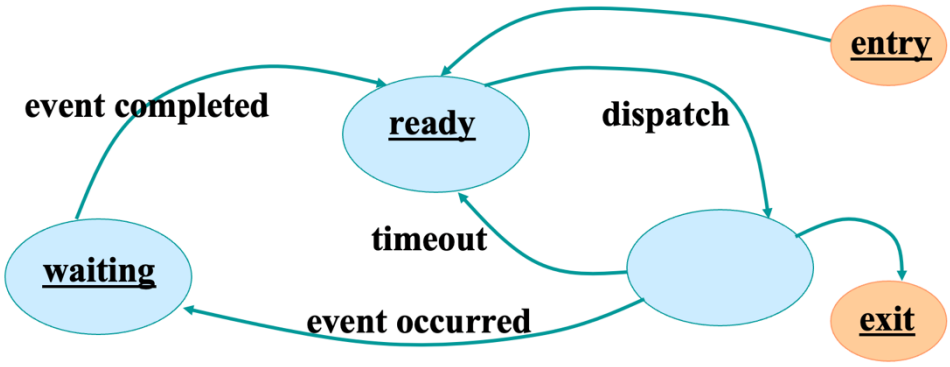
File System Management (1 mark) - *Organises and manages files and directories allowing users to create* ***OR*** *read* ***OR*** *write* ***OR*** *delete files* ***(1 mark) WHILE*** *ensuring data integrity and security through permissions and access control.* ***(1 mark)***

Device Management (1 mark) - *Interacts with hardware devices, providing device drivers for software applications to communicate with peripherals* ***(1 mark) LIKE*** *keyboards, mice, printers, and network adapters.* ***(1 mark)***

Security (1 mark) - *Implements security measures like user authentication* ***OR*** *access control* ***OR*** *encryption* ***OR*** *firewall configurations* ***(1 mark)*** *to protect the system from unauthorised access* ***OR*** *malware* ***OR*** *safeguarding sensitive data and resources.* ***(1 mark)***

Note: a description does not need to be exhaustive (the series of accepted answers are distinguished by ***OR / AND / LIKE / WHILE****)*

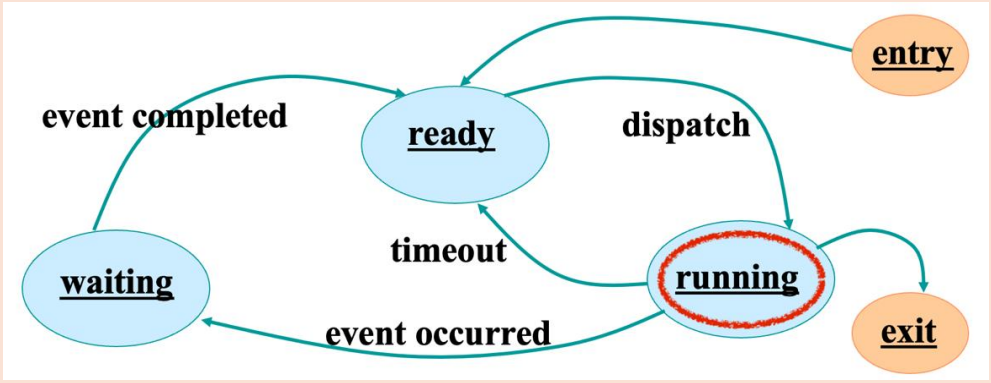
b) Identify the missing state as part of the process state diagram: **1**



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Mark Scheme

"Running" (1 mark)



Total 10 Marks

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Question 5

The Central Processing Unit (CPU) is the fundamental digital data processor in a computer system.

a) Identify the THREE (3) main functions of a CPU? **3**

Mark Scheme:

Fetch instruction from RAM ***(1 mark)***
Decode instruction ***(1 mark)***
Execute instruction ***(1 mark)***

b) Explain the purpose of an operand in a CPU? **2**

Mark Scheme:

To represent the data on which an operation is to be performed. ***(1 mark)*** *It holds the data needed for executing instructions such as arithmetic, logic and data manipulation operations.* ***(1 mark)***

c) Explain the purpose of an opcode in a CPU? Give ONE (1) example of an operation it performs. **3**

Mark Scheme:

To specify the operation that the CPU should perform. ***(1 mark)*** *It serves as a code that indicates the type of operation, such as addition OR subtraction OR data movement* ***(1 mark)****, enabling the CPU to execute the appropriate instruction* ***(1 mark)****.*

d) Before the CPU processes any data, it is recommended that data is checked via a parity checker. Outline how a parity checker works. **2**

Mark Scheme:

A parity checker for a CPU is a component or circuitry responsible for verifying the integrity of data transmitted to or stored in the CPU's memory ***(1 mark)****. It works by adding an extra parity bit to each byte of data, which is calculated based on the number of bits set to 1 in the byte (either even or odd parity).* ***(1 mark)***

Total 10 Marks

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Question 6

Data communications networking is an integral part of modern computer systems.

a) Identify THREE (3) main goals of successful networking of data? **3**

Mark Scheme:

Up to THREE (3) from the following:

Success – to allow data to be transferred from one device to its intended destination ***(1 mark)***
Speed – to transfer data from one device to another quickly ***(1 mark)***
Integrity – to ensure data remains intact/complete throughout the transmission ***(1 mark)***
Secure – to ensure data is transmitted securely without unauthorised intervention ***(1 mark)***

Answers may differ from the above suggestions. There may be alternative suitable answers provided by students

b) The internet is an example of a mode of data transmission. Consider the following statement: **3**

"The internet is based on a synchronous transmission system."

State whether the above statement is *true* or *false* and explain how a synchronous mode of transmission functions.

Mark Scheme:

False – the internet is based on an asynchronous mode of transmission. (1 mark)

Synchronous transmission relies on the transmitter and receiver being synchronised (1 mark). Synchronisation bits are transmitted by the sender to keep the receiver in sync (1 mark).

c) BUS and RING are two examples of LAN topologies. Name ONE (1) advantage and ONE (1) disadvantage of each topology. **4**

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Mark Scheme:

For BUS, any 1 advantage and any 1 disadvantage from:
Bus Advantages:
Cost-Effective** – cheap to install **(1 mark)
Simple to implement** – It's linear, so straightforward to install for a small network **(1 mark)
Scalable** – Its generally very easy to add devices to bus network as they just get added to the end **(1 mark)

Bus Disadvantages:
Network disruption** – Susceptible to complete failure if main cable (bus) fails **(1 mark)
Limited scalability** – Performance degrades as more devices are added **(1 mark)
Collisions** – Heavy traffic will increase collisions and congestion **(1 mark)
Not Secure** – Not secure by nature (easy to eavesdrop) **(1 mark)

For RING, any 1 advantage and any 1 disadvantage from:

Ring Advantages:
Equal Access** – Every device has equal access to network (prevent collisions) **(1 mark)
Performance** – Data transmission is orderly and predictable since each device has a chance to send data in a defined sequence **(1 mark)
Secure** – Secure by nature **(1 mark)

Ring Disadvantages:
Troubleshooting Difficulty** – Diagnosing issues is challenging due to failure in a single node or cable segment can disrupt entire network **(1 mark)
Scalability issues** – Adding or removing devices requires closing the network, making it less flexible for dynamic network environments **(1 mark)
Expensive** – Costs a lot to implement. **(1 mark)

Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.

Total 10 Marks

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Question 7

a) Micro-controllers are widely used in a number of different areas. Provide FOUR (4) examples of application areas where micro-controllers are used. **4**

Mark Scheme:

Any 4 from below to a maximum of 4 marks:
Automotive industry ***(1 mark)***
Security industry ***(1 mark)***
Domestic appliances ***(1 mark)***
Military equipment ***(1 mark)***
Scientific instruments ***(1 mark)***
Computer peripherals ***(1 mark)***
Entertainment industry ***(1 mark)***

b) Explain the difference between a polled interrupt and a vectored interrupt. **2**

Mark Scheme:

Polling: *CPU continuously checks interrupt status by polling a register or memory location* ***(1 mark).***

Vectored: *Each interrupt source is associated with a unique interrupt vector which automatically directs the CPU to the corresponding Interrupt Service Routine (ISR)* ***(1 mark)****.*

c) Real Time Operating Systems (RTOS) are specialised operating systems designed to handle real-time applications with precise timing and high reliability.

Outline FOUR (4) distinct characteristics of a RTOS's operation that you may find on an embedded system, such as a car's electronic control unit (ECU). **4**

Mark Scheme:

Any 4 from below to a maximum of 4 marks:
Task scheduling *- mention of pre-emptive OR priority-based scheduling* ***(1 mark)***

Fixed memory *- dynamic memory OR static memory* ***(1 mark)***

Deterministic timings *– mention of interrupts OR delays OR control OR context switching OR task creation* ***(1 mark)***

Task synchronisation (1 mark)

Reliability and Fault Tolerance *– mention of mechanisms for task monitoring OR fault recovery OR system diagnostics* ***(1 mark)***

Interrupt Handling *– mention of fast/efficient handling of real-time interrupts* ***(1 mark)***

Total 10 Marks

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Question 8

Reduced Instruction Set Computing (RISC) is a CPU architecture design philosophy that emphasises simplicity and efficiency of operation.

a) Identify THREE (3) features of RISC that allow it to operate simply and efficiently. **3**

Mark Scheme:

Any 3 from below to a maximum of 3 marks:
Large set of registers (32, 64, 128 or more) ***(1 mark)***
Uses registers for passing parameters to functions ***(1 mark)***
Uses registers to return values from functions ***(1 mark)***
Global variables stored in registers ***(1 mark)***
Low number of instructions (less than 100) ***(1 mark)***
Reduced number of addressing modes ***(1 mark)***
Fixed size instructions (64 / 128 bytes) ***(1 mark)***

b) Identify THREE (3) advantage of RISC when compared to Complex Instruction Set Computing (CISC). **3**

Mark Scheme:

Any 2 from below to a maximum of 2 marks:
Simplified Instruction Set ***(1 mark)***
Faster execution ***(1 mark)***
Compiler optimisation ***(1 mark)***
Improved performance ***(1 mark)***
Scalability and flexibility ***(1 mark)***
Reduced hardware complexity ***(1 mark)***
Better pipelining support ***(1 mark)***

c) Explain the role of context switching when two programs compete for the CPU time. **4**

Mark Scheme:

Context switching saves the context state ***(1 mark)*** *of PC Register* ***OR*** *Registers* ***OR*** *Status Register - SR* ***(1 mark)*** *of one program before yielding the CPU and switching to the other program* ***(1 mark)****. The other program's context state is then restored before it takes control of the CPU* ***(1 mark)****.*

Total 10 Marks

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Question 9

a) Scenario

A university library database keeps track of its books, members and the borrowing process. The structure of the database is as follows:

1. **Books:** Each book has a unique ISBN, a title, an author and a publication year.

2. **Members:** Each member has a unique member ID, a name, an email address and a membership start date.

Borrowing: Each borrowing transaction records the member who borrowed the book, the book borrowed, the date borrowed and the due date.

i) In relation to the given scenario, identify THREE (3) attributes that exist in the university library database. **3**

Mark Scheme:

Any TWO (2) from the following (or similarly worded)*: unique ISBN, book title, book author, book publication year, unique member ID, member name, member email address, member start date, date of borrowing, due date, member borrowing, borrowing transaction.*

(up to 2 marks)

ii) In relation to the given scenario, what Structured Query Language (SQL) command would be used to add the following information to the *Books* table? **1**

9780763707996, Computer Science Illuminated, Nell B. Dale, 2004

Mark Scheme:

INSERT or INSERT INTO (1 mark)

b) State is the primary purpose of Domain Name Service (DNS) in computer networking. **1**

Mark Scheme:

The DNS serves to translate human-readable domain names into IP addresses (or similar, 1 mark)

c) Some developers use virtual machines. Describe what a virtual machine is and outline THREE (3) of its purposes. **5**

Mark scheme:

A virtual machine simulates **(1 mark)** the instructions of a guest computer on a host computer (**1 mark)**. It is used to test different operating systems on a host operating system **(1 mark)**; to execute applications designed for one guest operating system on a host operating system **(1 mark)**; and to run multiple guest operating systems on a single host operating system **(1 mark)**.

Total 10 Marks

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Question 10

To enhance computer system's performance, advanced technologies have been developed, besides increasing CPU clock speeds.

a) A CPU, with its pipeline switched off, executes 1800 instructions in 5400 clock cycles. What is the average clock cycle per instruction executed? **1**

Mark Scheme:

*5400 / 3600 = **3** instructions per clock cycle (1 mark for correct answer)*

b) Explain the purpose of a CPU pipeline? **2**

Mark Scheme:

*The purpose of a CPU pipeline is to improve the efficiency and speed of instruction execution by overlapping the execution phases of multiple instructions. **(1 mark)** This allows the CPU to process several instructions simultaneously, increasing overall throughput and performance. **(1 mark)***

c) Explain how a multi-core CPU has enhanced performance over a single-core CPU. **2**

Mark scheme:

*In a multi-core CPU, there are more independent CPUs with their separate caches **(1 mark).** Each CPU can execute different programs' instructions at the same time. **(1 mark)***

d) Outline FIVE (5) characteristics of distributed computing? **5**

Mark scheme:

Any 5 from the following:
*Multiple computer systems working together to solve complex problems **(1 mark)***
*Combining multiple computer systems computational power **(1 mark)***
*Resource Sharing (both hardware and software) **(1 mark)***
*Remote process management **(1 mark)***
*Increased Reliability (or Fault Tolerance) **(1 mark)***
*Scalability (or Incremental Growth) **(1 mark)***
*Multiple platform interoperability **(1 mark)***

Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.

Total 10 Marks

End of paper

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Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO 1,2,3,4,5	Yes
2	LO 1	Yes
3	LO 3	Yes
4	LO 2	Yes
5	LO 4	Yes
6	LO 7	Yes
7	LO 1, 6	Yes
8	LO 4	Yes
9	LO 5	Yes
10	LO 1, 6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand and identify the main types and components of computer systems	None or inadequate description of computer systems and components	Some credible but incomplete attempt to demonstrate understanding of computer systems and components	Limited but satisfactory demonstration of understanding of computer systems and components	Good demonstration of understanding of computer systems and components	Excellent demonstration of understanding of computer systems and components
Describe the structure and role of modern operating systems	None or very basic description of components of operating systems	A very limited and basic description of components of operating systems	Limited and above basic description of components of operating systems	Good description of most components of operating systems	Excellent and full description of all components of operating systems
Understand and work with binary numbers and computer logic	No demonstration of working with binary numbers and logic operators	A basic demonstration of restricted range of working with binary numbers and logic operators	An above basic demonstration of working with restricted range of binary numbers and logic operators	A good demonstration of working with a wide range of binary numbers and logic operators	An excellent demonstration of working with full range of binary numbers and logic operators
Understand the essential structure of computer programs	None or trivial explanation of instruction sets, addressing modes and interrupts	Limited understanding and explanation of instruction sets, addressing modes and interrupts	Basic understanding and explanation of instruction sets, addressing modes and interrupts	Good understanding and explanation of instruction sets, addressing modes and interrupts	Excellent understanding and explanation of instruction sets, addressing modes and interrupts
Explain the hierarchy of computer	No or wrong explanation of computer software	Limited and very basic explanation of computer software	Basic but limited explanation of computer software	Good and competent explanation of computer	Excellent and complete explanation of computer

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software	hierarchy	hierarchy	hierarchy	software hierarchy	software hierarchy
Describe alternative computer systems	No or insufficient description of performance enhancing technologies	Limited and very basic description of performance enhancing technologies	Basic and partly satisfactory description of performance enhancing technologies	Good description of performance enhancing technologies	Excellent and complete description of performance enhancing technologies
Discuss the role of Data Communications and Networks in computer systems	No or very trivial discussion of data communications protocols and components	A very basic discussion of data communications protocols and components	Basic but credible discussion of data communications protocols and components	Good and wide-ranging discussion of data communications protocols and components	Excellent and complete discussion of data communications protocols and components

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